

Hanxi “Gary” Chen

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EDUCATION & AWARDS

Cornell University

Ithaca, NY • 2025 – Present

Ph.D. in Computer Science, Programming Languages

I work on end-to-end security for probabilistic and concurrent systems: designing adversary-aware program logics, exploring AI-assisted proof and invariant synthesis, and building security-preserving compilation frameworks.

Advisors: Prof. Alexandra Silva, Prof. Andrew C. Myers

University of Pennsylvania, triple degree

Philadelphia, PA • 2020 – 2025

M.S.E. in Computer and Information Science

B.S.E. in Computer Science, Mathematics minor

B.S. in Economics, The Wharton School

GPA: 3.97/4.00 • Honors: *Summa Cum Laude* • Awards: MCIT TA Award, CIS Faculty Appreciation Award

Advisors: Prof. Steve Zdancewic, Prof. Sanjeev Khanna

SELECTED PUBLICATIONS

Hanxi Chen, Noam Zilberstein, Andrew C. Myers, Alexandra Silva. [Oblivious Probabilistic Outcome Logic: Verifying Probabilistic Programs with an Oblivious Adversary](#). arXiv preprint, 2026. Presented at the Cornell PL Seminar (PLDG), Sep 2026.

Calvin Beck, **Hanxi Chen**, Steve Zdancewic. [Vellvm: Formalizing the Informal LLVM \(An Experience Report\)](#). NFM'25.

Calvin Beck, **Hanxi Chen**, Steve Zdancewic. [Vellvm: Formalizing the Informal](#). CoqPL'25.

Calvin Beck, Irene Yoon, **Hanxi Chen**, Yannick Zakowski, Steve Zdancewic. [A Two-Phase Infinite/Finite Low-Level Memory Model: Reconciling Integer–Pointer Casts, Finite Space, and undef at the LLVM IR Level of Abstraction](#). ICFP'24.

RELEVANT RESEARCH EXPERIENCE

Research Assistant, *Cornell PL Group*, advised by Prof. Alexandra Silva and Prof. Andrew C. Myers

Sep 2025 – Present

Cornell University, Computer Science

- Developed Oblivious Probabilistic Outcome Logic (opOL), a separation logic for probabilistic nondeterministic programs with oblivious adversaries, supporting compositional reasoning about probabilistic independence and unbounded loops.
- Mechanized opOL's metatheory in Lean 4 and verified case studies inspired by randomized caching and leader-election protocols.
- Co-developed a theory of type-confusion security for decentralized systems, where untrusted components misrepresent dynamic types; led its Lean mechanization and prove type safety for static and dynamic checks under ideal-world and real-world attacker models.

Research Mentor, *Cornell Math+AI*, advised by Prof. Daniel Halpern-Leistner

Jan 2026 – Present

Cornell University, Mathematics

- Mentored the Jan–Aug 2026 iteration of an autoformalization project, producing a proof-quality evaluation platform and MathCopilot.org, a LaTeX-to-Lean translation tools with interactive dependency graphs
- Also mentoring a second team building agentic framework for synthesizing invariants for program logics in software engineering.

Research Assistant, *PLClub*, advised by Prof. Steve Zdancewic

Jun 2021 – Aug 2025

University of Pennsylvania, Computer and Information Science

- A main contributor to the Verified LLVM Project (Vellvm), which develops formal semantics for LLVM IR in Rocq using interaction tree and coinduction; co-authored its ICFP 2024 paper, helping design a low-level memory model for optimization reasoning.
- Random Testing at IR Level (senior thesis): built a Rocq/OCaml differential-testing framework validating Vellvm against real LLVM, with QuickChick generators for UB-free programs; found previously unreported LLVM bugs.
- Developed Rocq infrastructure for permutation reasoning and list AC unification, then developed a Rocq formalization for polymorphic, multiplicative-exponential linear logic with cut elimination.
- Contributed to DeepSpec Software Foundations material presented at the Oregon Programming Languages Summer School 2024.

Research Assistant, *Wharton Research Scholars*, advised by Prof. Sanjeev Khanna

Jan 2023 – Dec 2024

University of Pennsylvania, The Wharton School

- Knapsack Problem with Precedence Constraints (Wharton Research Scholars thesis): designed pseudo-polynomial algorithms on structured poset classes – unions of three paths and one-directional bipartite posets with consecutive children.

SELECTED TECHNICAL PROJECTS

RISC-V Pipelined Processor – *SystemVerilog, Python, FPGA*

Feb 2025 – May 2025

- Designed a RISC-V pipelined processor with AXI-Lite caching, multi-cycle units, branch prediction, and memory-mapped I/O.
- Built verification testbenches, optimized LUT usage and timing behavior, and synthesized the design to FPGA to run compiled programs.

Oat Compiler – *OCaml, C, LLVM IR, x86, Clang, Dune*

Feb 2024 – May 2024

- Implemented a compiler for Oat, a C-like language with pointers and structured data, including lexing, parsing, type checking, frontend lowering, and backend code generation to LLVM IR and x86.
- Added static analyses and optimizations including mem2reg and dead-code elimination, validated against thousands of test cases.

SKILLS, AREAS & TEACHING

Proof assistants & verification: Lean 4, Rocq, Z3, property-based testing (QuickCheck).

Languages & tools: OCaml, Haskell, Python, C/C++, Java, Rust, PyTorch, SystemVerilog, LLVM, Clang, Assembly, RISC-V, Git, Docker.

Areas: formal verification, program logics, probabilistic programming, concurrency, compilers, information flow.

Teaching: 12 appointments, 5 as Head TA – formal verification, programming languages, compilers, algorithms, network theory.